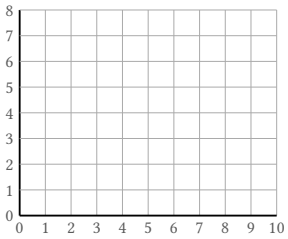
Regions

Name: _____ Date: _____

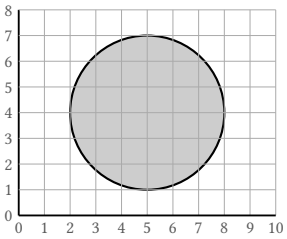
A drawing program, a map, or a game has to answer one question constantly: is this point inside this shape? Here every shape is a `Region` that can answer it. A rectangle's `corner` is its bottom-left corner.

```
record Point(double x, double y) {}
interface Region { boolean contains(Point p); }
record RectRegion(Point corner, double width, double height) implements Region {
  public boolean contains(Point p) { ... }
}
record CircRegion(Point center, double radius) implements Region {
  public boolean contains(Point p) { ... }
}
```

Q1. Shade `new RectRegion(new Point(1, 2), 4, 3)`.

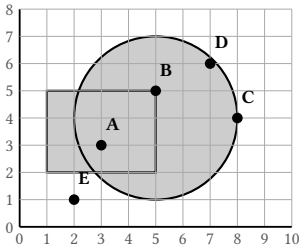


Q2. Construct this region.



From here on, `box` and `ball` are the regions from Q1 and Q2.

Q3. Which points does each region contain? Decide by looking. Leave the last column for later.



point	`box.contains`	`ball.contains`	`new Union(box, ball).contains`
A (3, 3)			
B (5, 5)			
C (8, 4)			
D (7, 6)			
E (2, 1)			

Q4. Two of the points sit exactly on an edge. Which? Does a region contain its own edge? Decide; your code below makes it official.

Q5. Write the two `contains` methods.

Q6. A point is in `new Union(box, ball)` when it is in either one. Finish this record, then fill in the last column of Q3.

```
record Union(RectRegion a, CircRegion b) {
  public boolean contains(Point p) {

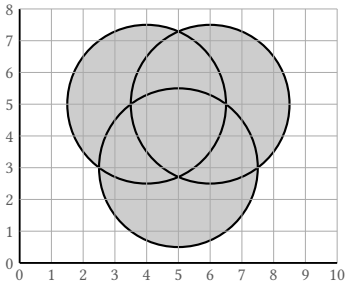
    return                                          ;
  }
}
```

Here are three more regions:

```
Region one = new CircRegion(new Point(4, 5), 2.5);
Region two = new CircRegion(new Point(6, 5), 2.5);
Region three = new CircRegion(new Point(5, 3), 2.5);
```

Q7. Can the `Union` from Q6 combine `one` and `two`? What does Java say? Rewrite the first line of `Union` so it can.

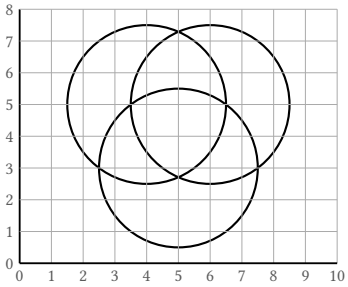
Q8. Construct this region, from `one`, `two`, and `three`. A `Union` holds only two regions. What else has to change in `Union` for your answer to compile?

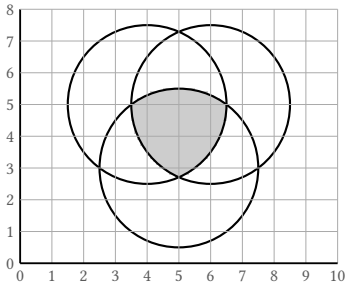


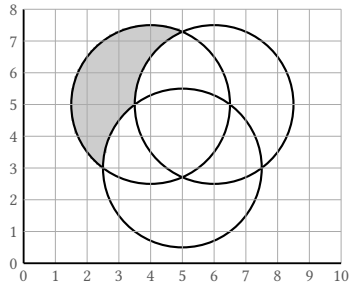
Q9. A point is in `new Intersect(a, b)` when it is in both `a` and `b`. Write the whole `Intersect` record.

Q10. Shade the first region. Construct the other two.

```
new Intersect(new Union(one, two),
  three)
```







Q11. The last one needs a new kind of region, `Negate`: a point is in `new Negate(r)` when it is **not** in `r`. Write the `Negate` record.